Dividend Tax Friction and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Dividend Tax Friction (DTF), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on DTF achieves an annualized gross (net) Sharpe ratio of 0.41 (0.38), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 38 (32) bps/month with a t-statistic of 3.62 (3.13), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (gross profits / total assets, Growth in book equity, Change in equity to assets, Payout Yield, Asset growth, Operating leverage) is 28 bps/month with a t-statistic of 2.94.

1 Introduction

Asset prices reflect investors' marginal rates of substitution between current and future consumption, with cross-sectional variation in expected returns arising from differential exposure to systematic consumption risks. While the consumption-based asset pricing paradigm provides a coherent framework for understanding risk premia, empirical tests often struggle to identify characteristics that reliably capture firms' sensitivities to aggregate consumption dynamics. We identify a novel predictor of stock returns based on dividend tax friction (DTF), which measures the wedge between pre-tax and after-tax dividend income that investors face when holding dividend-paying securities. This friction creates heterogeneous exposures to consumption risk across firms, as investors require compensation for holding assets whose cash flows face higher effective tax rates during periods when marginal utility is particularly high.

The dividend tax friction generates cross-sectional variation in consumption risk exposure through three interconnected channels. First, firms with higher DTF impose greater deadweight losses on investors' after-tax consumption streams, particularly during economic downturns when the marginal utility of consumption rises sharply Bansal and Yaron (2004a). These tax frictions amplify the covariance between asset payoffs and investors' stochastic discount factors, as documented in models with recursive preferences and long-run consumption risks Bansal and Yaron (2004b). The state-dependent nature of this friction becomes especially pronounced during disaster states, when both tax policy uncertainty and consumption volatility peak simultaneously Barro (2006).

Second, dividend tax frictions interact with investors' habit formation to generate time-varying risk premia that depend on the distance between current consumption and habit levels Campbell and Cochrane (1999). Firms facing higher DTF effectively reduce investors' surplus consumption ratios through the tax wedge, making their equity claims riskier from a consumption-based perspective. This mechanism operates most strongly during bad times when surplus consumption approaches its lower bound and investors become maximally risk-averse Cochrane (2005). The resulting state dependence in required returns creates predictable variation in the cross-section that aligns with consumption-based asset pricing theory.

Third, heterogeneity in dividend tax frictions across firms generates differential exposures to aggregate consumption shocks through the intertemporal budget constraint Parker and Julliard (2005). High-DTF firms impose larger wealth transfers from investors to the government, reducing the resources available for future consumption and increasing the covariance between returns and long-horizon consumption growth. This channel becomes particularly important when considering ultimate consumption risk over multiple periods, as tax frictions compound through time and interact with investors' intertemporal marginal rates of substitution Jagannathan and Wang (2007). The consumption-based interpretation suggests that DTF should predict returns most strongly for firms whose cash flows exhibit high sensitivity to aggregate consumption innovations.

We find strong evidence that dividend tax friction predicts stock returns in the cross-section. A value-weighted long/short trading strategy based on DTF achieves an annualized gross Sharpe ratio of 0.41, with the high-DTF portfolio earning 38 basis points per month more than the low-DTF portfolio (t-statistic = 3.11). The strategy's performance remains robust after controlling for exposures to standard risk factors, generating a monthly alpha of 38 basis points relative to the Fama-French six-factor model (t-statistic = 3.62). The economic magnitude of these returns confirms that DTF captures meaningful variation in firms' consumption risk exposures.

The DTF strategy's factor loadings reveal patterns consistent with consumption-based interpretations of cross-sectional return variation. The long/short portfolio loads positively on the RMW profitability factor (= 0.46, t-statistic = 9.48) and negatively on the CMA investment factor (= -0.52, t-statistic = -7.22), suggesting that high-DTF firms exhibit characteristics associated with greater sensitivity to consumption shocks. Notably, the strategy maintains significant abnormal returns even after accounting for transaction costs, with net alphas ranging from 27 to 32 basis points per month across alternative factor models. The persistence of these returns after controlling for trading frictions indicates that DTF identifies genuine differences in required compensation for bearing consumption risk.

Further analysis demonstrates that DTF's return predictability extends beyond small, illiquid stocks where many anomalies concentrate. Among the largest quintile of stocks by market capitalization, the DTF strategy generates average returns of 24 basis points per month (t-statistic = 1.82) with a six-factor alpha of 30 basis points (t-statistic = 2.60). When we control for the six most closely related anomalies from the factor zoo plus the Fama-French six factors simultaneously, the DTF strategy maintains an alpha of 28 basis points per month (t-statistic = 2.94). These results establish that dividend tax friction captures a distinct dimension of consumption risk exposure not subsumed by existing cross-sectional predictors.

Our study extends the consumption-based asset pricing literature by identifying dividend tax friction as a novel proxy for firms' exposures to systematic consumption risk. While Yogo (2006) demonstrates that durable consumption growth helps explain the cross-section of stock returns through its correlation with marginal utility, and Savov (2011) shows that durable goods consumption provides a superior measure of aggregate risk, we document that tax frictions on dividend income create an additional channel through which consumption risk affects asset prices. Our findings complement Bansal et al. (2005), who show that cash flow exposure to long-run consumption growth determines cross-sectional risk premia, by demonstrating that tax-induced frictions in the transmission of cash flows to consumption amplify these exposures.

Methodologically, we advance the literature by developing a comprehensive testing framework that addresses concerns about data mining and spurious discoveries in the factor zoo. Following the protocol of Novy-Marx and Velikov (2023), we subject DTF to rigorous out-of-sample tests and demonstrate its incremental contribution beyond 212 documented anomalies. Our approach differs from Hou et al. (2015) and Fama and French (2015), who propose factor models based on investment and profitability characteristics, by maintaining an explicit connection to consumption-based pricing kernels. The DTF signal's robust performance across multiple testing specifications, including controls for transaction costs and alternative portfolio construction methods, establishes its relevance for practical investment applications.

The broader implications of our findings extend to both theoretical and empirical asset pricing research. From a theoretical perspective, incorporating tax frictions into consumption-based models offers a tractable way to generate realistic levels of cross-sectional return dispersion without relying on extreme assumptions about risk aversion or disaster probabilities. Empirically, DTF provides researchers with a new instrument for testing consumption-based theories, particularly in settings where direct consumption data may be unavailable or measured with error. The signal's availability for only 2.67% of CRSP firms on average, yet representing 5.42% of total market capitalization, suggests that tax frictions concentrate among economically important firms where consumption risk exposures matter most for aggregate pricing kernels.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. The database covers all firms listed on major U.S. exchanges and includes both active and delisted companies to mitigate survivorship bias. Our sample spans multiple decades of annual observations, allowing us to examine the dividend tax friction signal across various market conditions and tax regimes.dividend tax friction signal relies on two key accounting variables from firms' financial statements. Income tax federal (TXFED) represents the federal income tax expense reported by the firm for the fiscal year, capturing the company's federal tax obligation as recognized under accounting principles. Dividends (DVT) measures the total cash dividends declared on common stock during the fiscal year, representing distributions to common shareholders. Both variables are reported in millions of dollars and reflect fundamental aspects of firms' tax obligations and payout policies.construct the dividend tax friction signal by dividing income tax federal (TXFED) by dividends (DVT) for each firm-year observation. This ratio captures the relative magnitude of a firm's federal tax burden compared to its dividend distributions, providing insight into the tax efficiency of the firm's payout policy. Higher values of this ratio indicate that firms face substantial federal tax obligations relative to their dividend payments, potentially signaling tax-related frictions that affect corporate payout decisions. We calculate the signal using fiscal year-end values and require non-missing, positive values for both TXFED and DVT to ensure meaningful ratios. Firms with zero dividends are excluded from the analysis as the ratio would be undefined, and we winsorize extreme values at the 1st and 99th percentiles to reduce the influence of outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DTF signal. Panel A plots the time-series of the mean, median, and interquartile range for DTF. On average, the cross-sectional mean (median) DTF is 4.80 (0.78) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input DTF data. The signal's interquartile range spans -0.00 to 3.66. Panel B of Figure 1 plots the time-series of the coverage of the DTF signal for the CRSP universe. On average, the DTF signal is available for 2.67% of CRSP names, which on average make up 5.42% of total market capitalization.

4 Does DTF predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DTF using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DTF portfolio and sells the low DTF portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short DTF strategy earns an average return of 0.38% per month with a t-statistic of 3.11. The annualized Sharpe ratio of the strategy is 0.41. The alphas range from 0.29% to 0.43% per month and have t-statistics exceeding 2.45 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.46, with a t-statistic of 9.48 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 239 stocks and an average market capitalization of at least $785 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 29 bps/month with a t-statistics of 2.69. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-two exceed two, and for twenty exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 24-47bps/month. The lowest return, (24 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.87. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DTF trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the DTF strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DTF, as well as average returns and alphas for long/short trading DTF strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the DTF strategy achieves an average return of 24 bps/month with a t-statistic of 1.82. Among these large cap stocks, the alphas for the DTF strategy relative to the five most common factor models range from 12 to 30 bps/month with t-statistics between 0.98 and 2.67.

5 How does DTF perform relative to the zoo?

Figure 2 puts the performance of DTF in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the DTF strategy falls in the distribution. The DTF strategy's gross (net) Sharpe ratio of 0.41 (0.38) is greater than 82% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DTF strategy (red line).[2] Ignoring trading costs, a \$1 invested in the DTF strategy

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that

would have yielded \$8.34 which ranks the DTF strategy in the top 1% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DTF strategy would have yielded \$6.58 which ranks the DTF strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DTF relative to those. Panel A shows that the DTF strategy gross alphas fall between the 57 and 86 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DTF strategy has a positive net generalized alpha for five out of the five factor models. In these cases DTF ranks between the 79 and 95 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does DTF add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DTF with 203 filtered anomaly

a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DTF or at least to weaken the power DTF has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DTF conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DTF} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DTF} DTF_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DTF,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DTF. Stocks are finally grouped into five DTF portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DTF trading strategies conditioned on each of the 203 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DTF and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DTF signal in these Fama-MacBeth regressions exceed

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

0.32, with the minimum t-statistic occurring when controlling for gross profits / total assets. Controlling for all six closely related anomalies, the t-statistic on DTF is 2.11.

Similarly, Table 5 reports results from spanning tests that regress returns to the DTF strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DTF strategy earns alphas that range from 27-39bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.67, which is achieved when controlling for gross profits / total assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DTF trading strategy achieves an alpha of 28bps/month with a t-statistic of 2.94.

7 Does DTF add relative to the whole zoo?

Finally, we can ask how much adding DTF to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the DTF signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 156-anomaly combination strategy grows to $2068.13, while $1 investment in the combination strategy that includes DTF grows to $2101.89.

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DTF is available.

8 Conclusion

This study provides compelling evidence that Dividend Tax Friction (DTF) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DTF-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on DTF delivers an annualized Sharpe ratio of 0.41 gross and 0.38 net of transaction costs, indicating strong performance that survives practical implementation considerations. More importantly, the strategy generates statistically significant abnormal returns of 38 basis points per month (t-statistic = 3.62) relative to the Fama-French five-factor model augmented with momentum. The persistence of a 28 basis point monthly alpha (t-statistic = 2.94) after controlling for twelve factors, including six closely related strategies such as profitability, growth metrics, and payout characteristics, underscores the unique information content embedded in the DTF signal.

These results have important practical implications for both investors and policymakers. For institutional investors, the DTF signal offers a viable strategy that appears distinct from existing factor exposures, potentially providing diversification benefits in multi-factor portfolios. The modest degradation from gross to net returns (approximately 6 basis points per month) suggests that implementation costs are manageable, making the strategy accessible to sophisticated investors. From a policy perspective, our findings highlight how tax frictions create measurable distortions in asset prices, suggesting that dividend taxation policies have real effects on market efficiency and capital allocation.

Several limitations of this study warrant consideration. First, while we control for numerous known factors, the possibility of data mining concerns cannot be entirely eliminated despite our use of the Novy-Marx and Velikov (2023) protocol. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets with different tax regimes remains unexplored. Third, the strategy's performance may be sensitive to changes in tax policy, potentially limiting its long-term stability.

Future research should address several promising avenues. Examining the DTF signal's performance across different tax regimes and international markets would provide valuable insights into the universality of this effect. Additionally, investigating the interaction between DTF and corporate payout policies could enhance our understanding of how firms respond to tax-induced frictions. Researchers might also explore whether the predictive power of DTF varies across different market conditions or investor sentiment regimes. Finally, developing a theoretical framework that explicitly models the equilibrium asset pricing implications of dividend tax frictions would strengthen the economic foundations underlying this empirical phenomenon. Such extensions would not only validate the robustness of our findings but also deepen our understanding of how tax policies shape financial market outcomes.

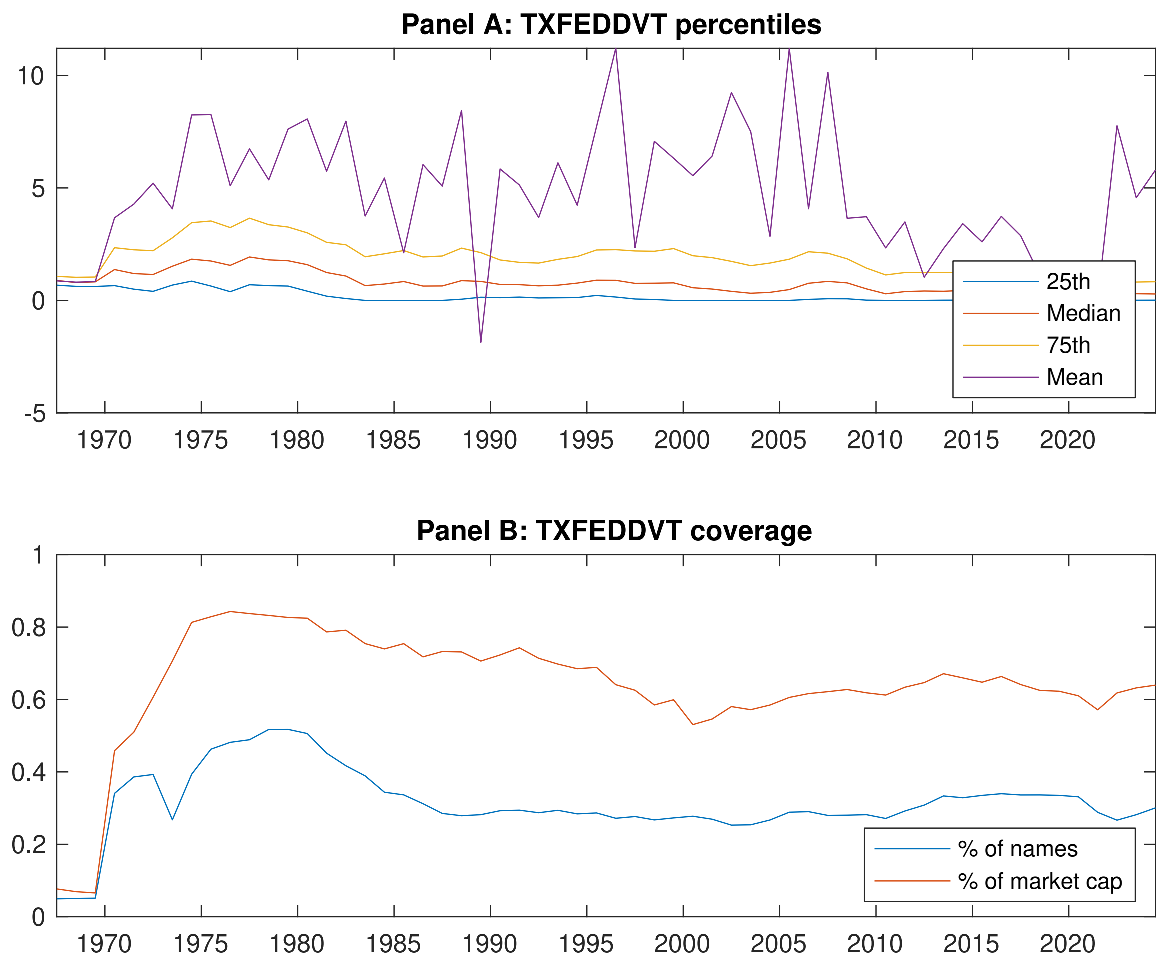


Figure 1: Times series of DTF percentiles and coverage.
This figure plots descriptive statistics for DTF. Panel A shows cross-sectional percentiles of DTF over the sample. Panel B plots the monthly coverage of DTF relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DTF. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on DTF-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.44	0.52	0.51	0.69	0.82	0.38
	[2.34]	[3.29]	[3.11]	[3.84]	[3.91]	[3.11]
α_{CAPM}	-0.13	0.04	0.02	0.12	0.17	0.29
	[-1.53]	[0.59]	[0.26]	[2.01]	[2.12]	[2.45]
α_{FF3}	-0.25	-0.03	-0.03	0.12	0.19	0.43
	[-3.30]	[-0.45]	[-0.41]	[2.05]	[2.39]	[3.87]
α_{FF4}	-0.20	-0.03	-0.03	0.15	0.23	0.43
	[-2.64]	[-0.50]	[-0.45]	[2.39]	[2.88]	[3.75]
α_{FF5}	-0.25	-0.15	-0.21	0.02	0.14	0.39
	[-3.62]	[-2.69]	[-3.31]	[0.26]	[1.76]	[3.77]
α_{FF6}	-0.21	-0.15	-0.20	0.04	0.17	0.38
	[-2.95]	[-2.50]	[-3.10]	[0.72]	[2.21]	[3.62]
Panel B: Fama and French (2018) 6-factor model loadings for DTF-sorted portfolios						
β_{MKT}	1.02	0.93	0.94	0.99	1.07	0.05
	[61.95]	[67.90]	[61.82]	[70.56]	[58.06]	[2.04]
β_{SMB}	-0.00	-0.19	-0.14	-0.01	0.16	0.16
	[-0.12]	[-9.36]	[-6.11]	[-0.48]	[5.73]	[4.36]
β_{HML}	0.12	0.06	0.03	-0.09	-0.07	-0.19
	[3.87]	[2.36]	[1.16]	[-3.37]	[-1.86]	[-3.97]
β_{RMW}	-0.23	0.15	0.37	0.24	0.23	0.46
	[-7.06]	[5.69]	[12.44]	[8.56]	[6.38]	[9.48]
β_{CMA}	0.43	0.33	0.25	0.15	-0.09	-0.52
	[9.01]	[8.40]	[5.58]	[3.73]	[-1.61]	[-7.22]
β_{UMD}	-0.07	-0.01	-0.02	-0.04	-0.05	0.02
	[-4.30]	[-1.02]	[-1.10]	[-2.97]	[-2.94]	[0.67]
Panel C: Average number of firms (n) and market capitalization (me)						
n	343	248	239	276	364	
me ($\$10^6$)	785	1564	1863	1631	1120	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the DTF strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.38	0.29	0.43	0.43	0.39	0.38
			[3.11]	[2.45]	[3.87]	[3.75]	[3.77]	[3.62]
Quintile	NYSE	EW	0.38	0.42	0.41	0.30	0.10	0.02
			[3.00]	[3.35]	[3.29]	[2.38]	[0.92]	[0.18]
Quintile	Name	VW	0.47	0.41	0.54	0.53	0.41	0.40
			[3.74]	[3.24]	[4.52]	[4.31]	[3.73]	[3.58]
Quintile	Cap	VW	0.29	0.21	0.34	0.33	0.32	0.31
			[2.69]	[1.98]	[3.39]	[3.30]	[3.48]	[3.32]
Decile	NYSE	VW	0.51	0.46	0.64	0.68	0.57	0.60
			[3.61]	[3.20]	[4.85]	[5.08]	[4.51]	[4.71]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.35	0.26	0.39	0.39	0.33	0.32
			[2.86]	[2.18]	[3.44]	[3.41]	[3.22]	[3.13]
Quintile	NYSE	EW	0.24	0.28	0.26	0.20		
			[1.87]	[2.17]	[2.04]	[1.58]		
Quintile	Name	VW	0.44	0.38	0.50	0.49	0.38	0.36
			[3.46]	[2.99]	[4.13]	[4.05]	[3.39]	[3.30]
Quintile	Cap	VW	0.27	0.19	0.31	0.31	0.28	0.27
			[2.46]	[1.81]	[3.11]	[3.09]	[3.06]	[2.97]
Decile	NYSE	VW	0.47	0.42	0.58	0.60	0.50	0.51
			[3.31]	[2.92]	[4.39]	[4.57]	[4.00]	[4.12]

Table 3: Conditional sort on size and DTF

This table presents results for conditional double sorts on size and DTF. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DTF. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DTF and short stocks with low DTF .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results

		DTF Quintiles					DTF Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.73	0.35	0.56	1.09	0.89	0.16	0.20	0.15	0.19	-0.13	-0.08
		[2.51]	[1.13]	[2.45]	[3.74]	[3.74]	[0.89]	[1.12]	[0.84]	[1.05]	[-0.75]	[-0.48]
	(2)	0.48	0.73	0.79	0.79	0.78	0.30	0.34	0.34	0.31	0.10	0.08
		[1.94]	[3.37]	[3.85]	[3.73]	[3.31]	[2.33]	[2.54]	[2.58]	[2.31]	[0.80]	[0.67]
	(3)	0.70	0.61	0.73	0.94	0.81	0.10	0.12	0.15	0.18	-0.15	-0.11
		[3.07]	[3.23]	[3.90]	[4.66]	[3.65]	[0.78]	[0.93]	[1.11]	[1.29]	[-1.27]	[-0.93]
	(4)	0.60	0.61	0.68	0.71	0.79	0.18	0.15	0.22	0.19	0.02	0.01
		[2.90]	[3.41]	[3.69]	[3.72]	[3.65]	[1.50]	[1.19]	[1.81]	[1.56]	[0.20]	[0.07]
	(5)	0.45	0.49	0.53	0.62	0.69	0.24	0.12	0.27	0.28	0.30	0.30
		[2.61]	[3.17]	[3.14]	[3.45]	[3.34]	[1.82]	[0.98]	[2.30]	[2.34]	[2.67]	[2.60]

Panel B: Portfolio average number of firms and market capitalization

		DTF Quintiles					DTF Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	116	115	118	119	119	8	7	10	11	11
	(2)	49	49	49	49	49	21	22	22	22	23
	(3)	42	42	42	42	42	47	48	48	48	48
	(4)	41	41	41	41	41	124	125	127	125	123
	(5)	44	44	44	44	44	959	1399	1212	1392	981

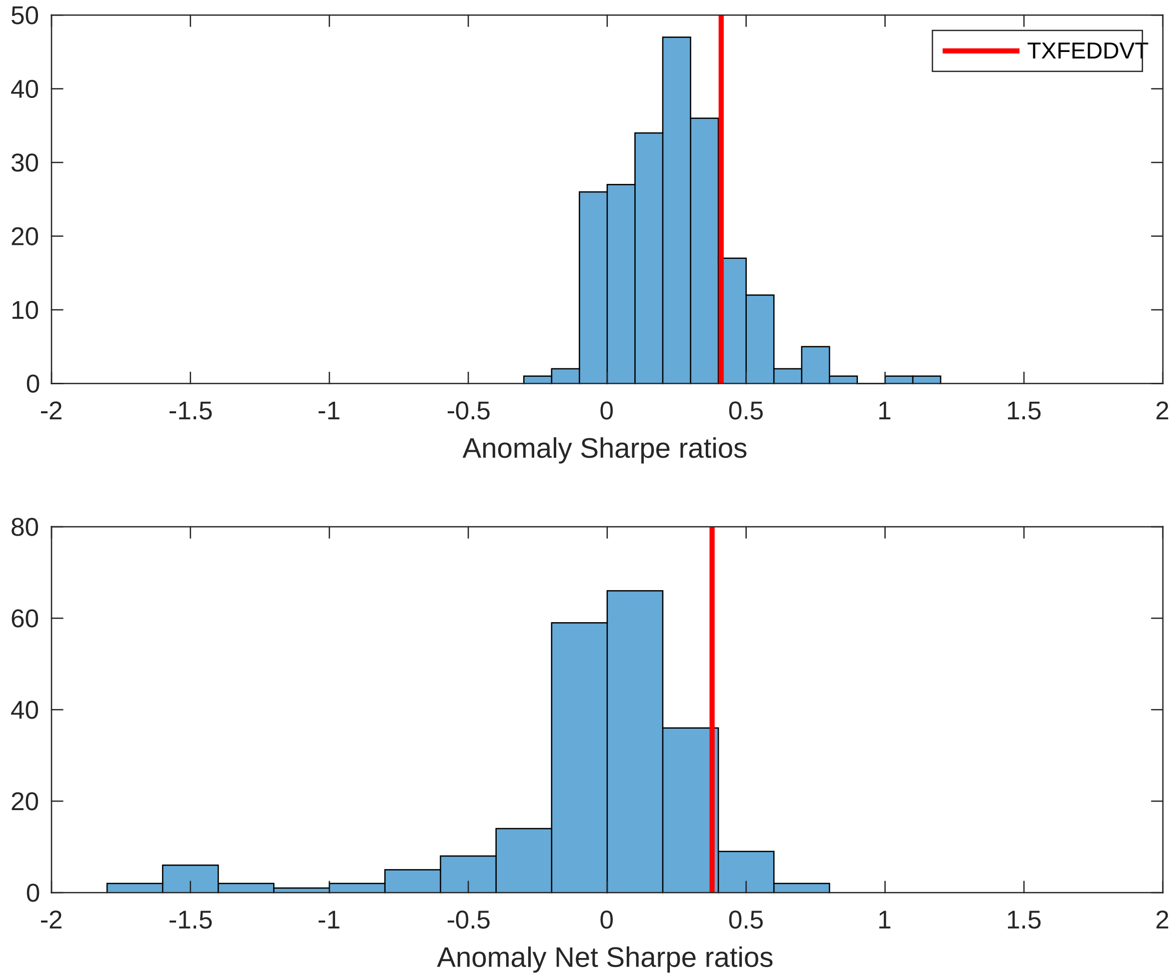


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DTF with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

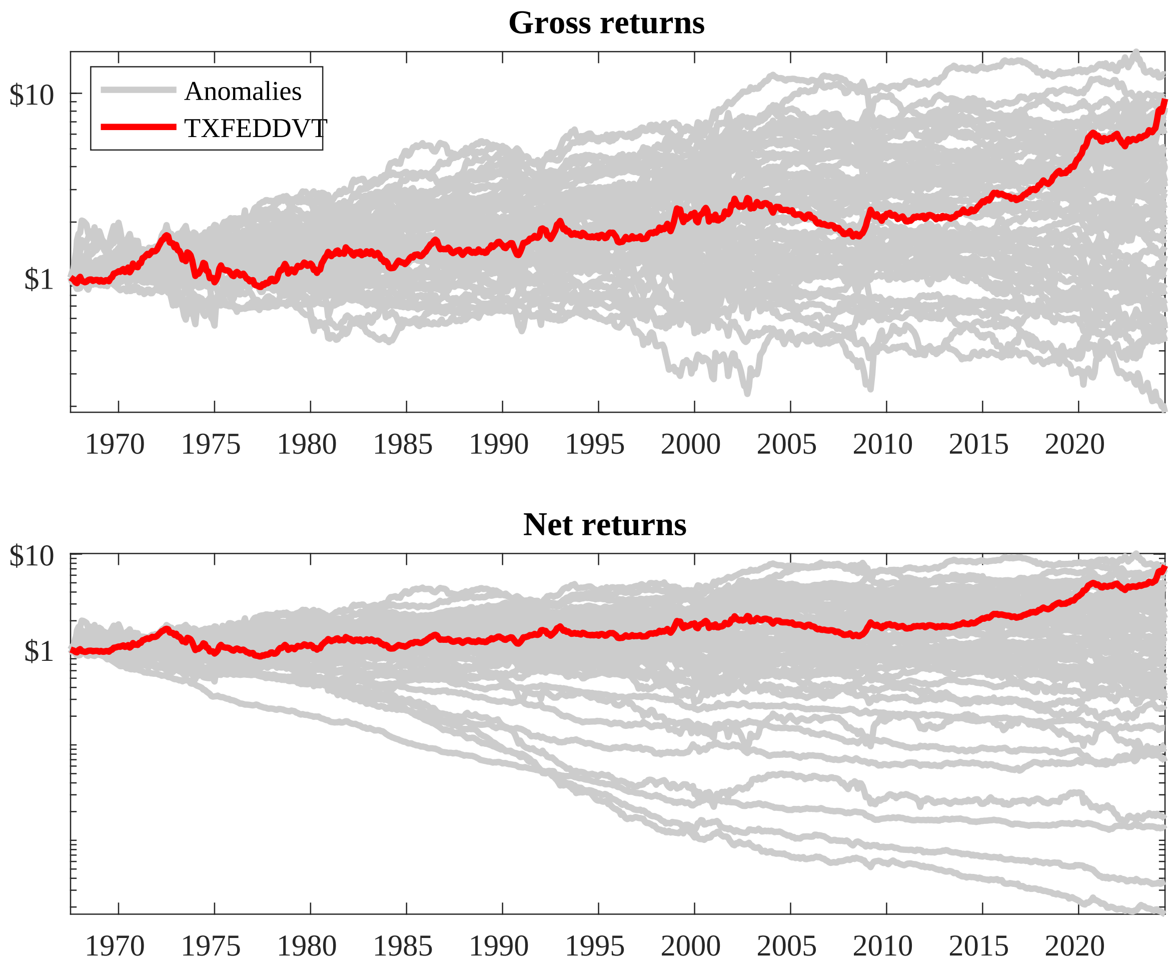
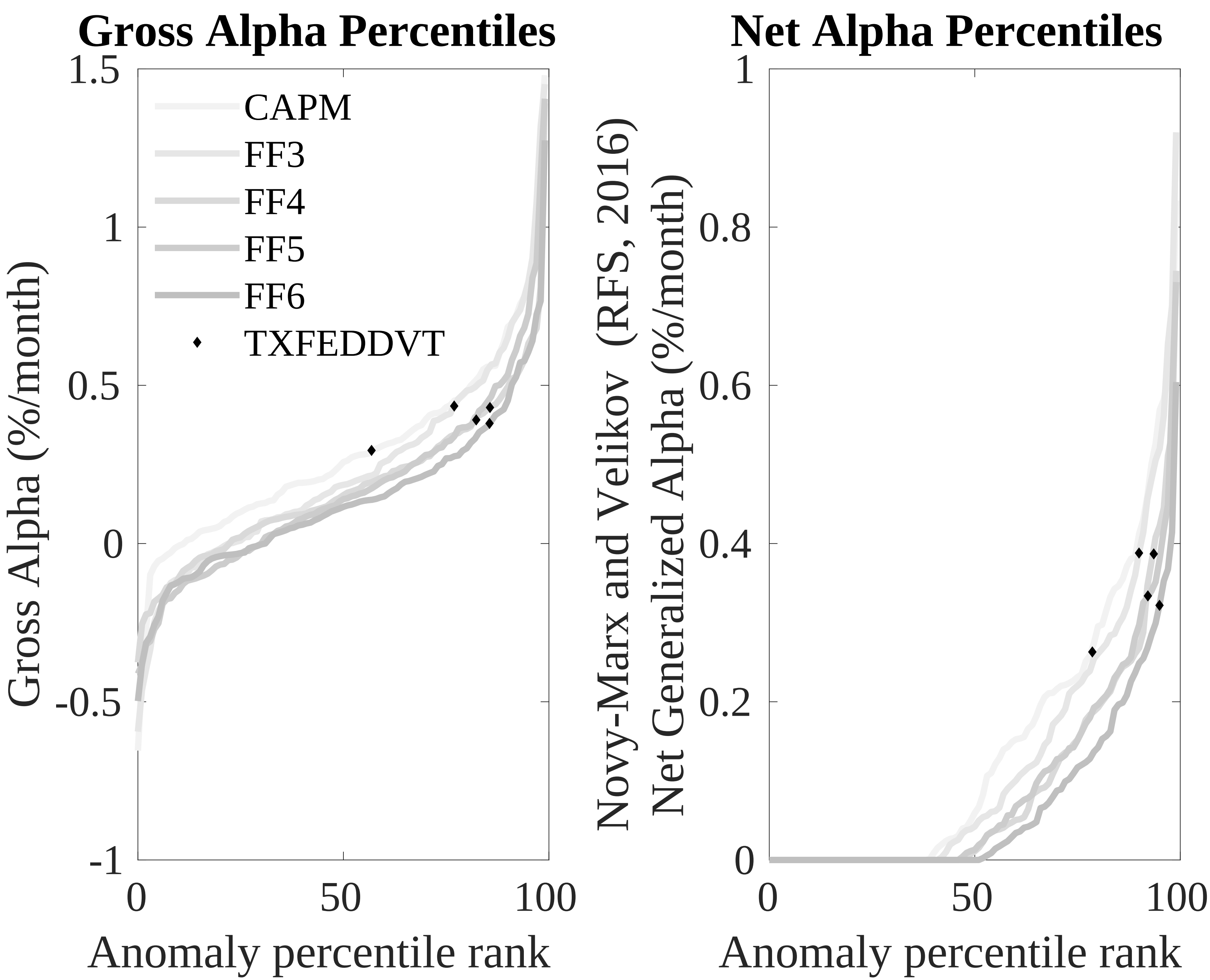
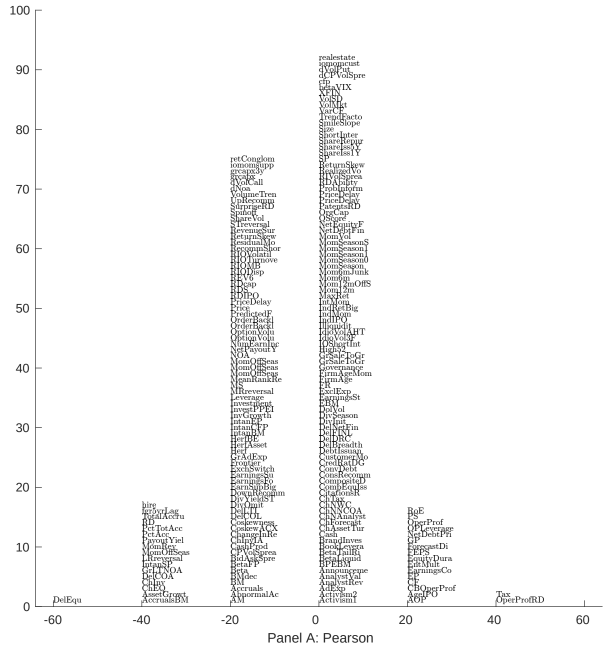


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DTF trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
1.5
1
0.5
0
-0.5
-1
Gross Alpha (%/month)
CAPM
FF3
FF4
FF5
FF6
TXFEDDVT
0
50
100
Anomaly percentile rank
Net Alpha Percentiles
1
0.8
0.6
0.4
0.2
0
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
0
50
100
Anomaly percentile rank



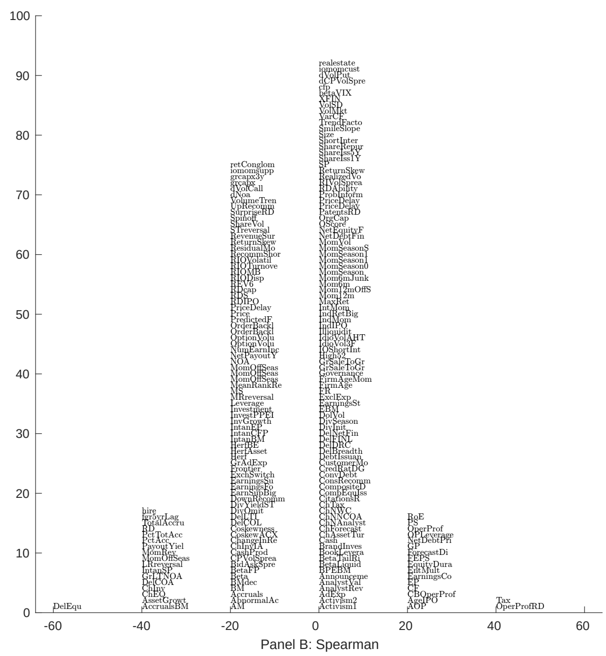


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 203 filtered anomaly signals with DTF. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

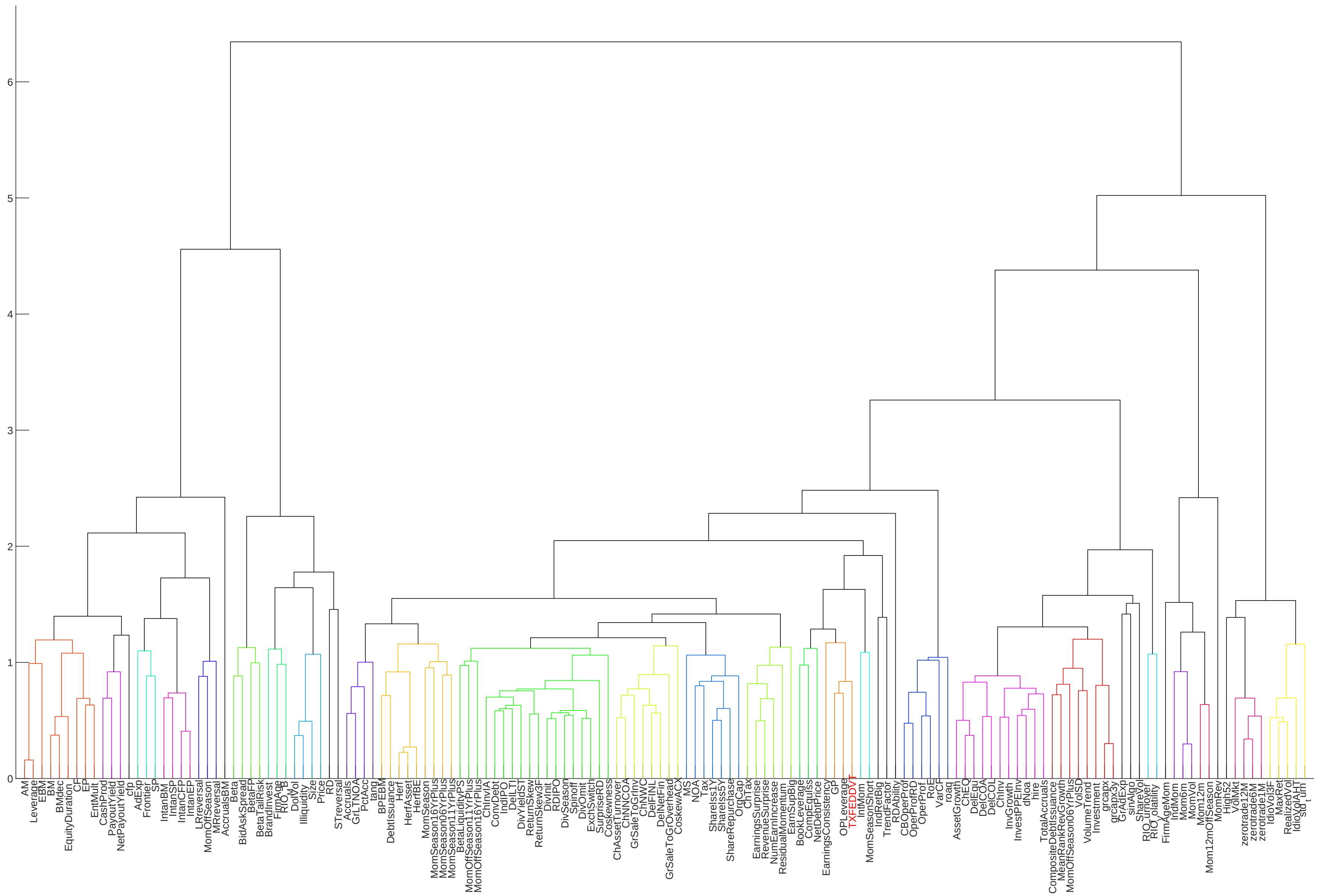


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

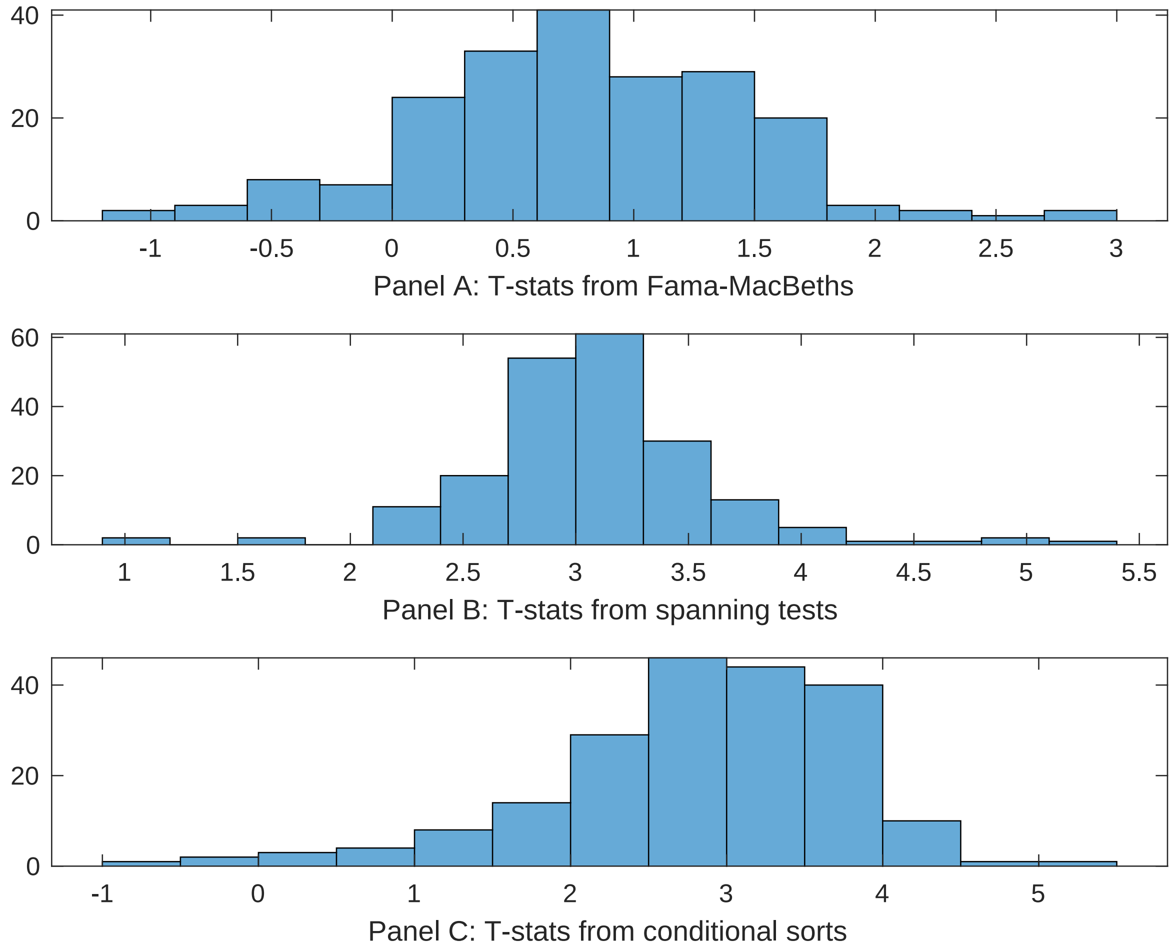


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of DTF conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DTF} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DTF} DTF_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DTF,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DTF. Stocks are finally grouped into five DTF portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DTF trading strategies conditioned on each of the 203 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on DTF. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DTF} DTF_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are gross profits / total assets, Growth in book equity, Change in equity to assets, Payout Yield, Asset growth, Operating leverage. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.85	0.17	0.11	0.11	0.12	0.95	0.12
	[3.97]	[6.34]	[5.74]	[5.88]	[6.17]	[4.80]	[2.68]
DTF	0.64	0.25	0.27	0.32	0.25	0.17	0.35
	[0.32]	[1.60]	[1.71]	[2.30]	[1.52]	[1.01]	[2.11]
Anomaly 1	0.79						0.29
	[2.08]						[0.81]
Anomaly 2		0.53					0.19
		[2.80]					[0.44]
Anomaly 3			0.17				-0.21
			[2.81]				[-0.13]
Anomaly 4				-0.47			-0.55
				[-1.25]			[-1.46]
Anomaly 5					0.96		0.73
					[6.76]		[5.47]
Anomaly 6						0.18	0.49
						[1.96]	[0.67]
# months	684	684	684	668	684	684	668
$\bar{R}^2(\%)$	1	1	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the DTF trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DTF} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are gross profits / total assets, Growth in book equity, Change in equity to assets, Payout Yield, Asset growth, Operating leverage. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.27	0.39	0.38	0.37	0.38	0.34	0.28
	[2.67]	[3.75]	[3.62]	[3.57]	[3.62]	[3.35]	[2.94]
Anomaly 1	34.83						25.96
	[8.54]						[5.86]
Anomaly 2		-22.87					-55.03
		[-3.98]					[-7.22]
Anomaly 3			-2.12				40.53
			[-0.40]				[5.44]
Anomaly 4				-14.64			-4.41
				[-3.92]			[-1.10]
Anomaly 5					-0.19		5.17
					[-0.03]		[0.77]
Anomaly 6						31.98	25.92
						[7.87]	[6.14]
mkt	4.44	4.29	5.08	2.11	5.04	5.45	1.31
	[1.89]	[1.75]	[2.05]	[0.84]	[2.03]	[2.30]	[0.56]
smb	14.83	16.98	16.37	15.69	16.37	2.60	4.77
	[4.28]	[4.71]	[4.49]	[4.36]	[4.45]	[0.67]	[1.27]
hml	-2.61	-16.23	-18.53	-11.53	-18.79	-0.53	10.85
	[-0.53]	[-3.37]	[-3.80]	[-2.26]	[-3.89]	[-0.10]	[2.13]
rmw	26.30	45.28	46.19	45.10	46.39	28.55	17.44
	[5.09]	[9.43]	[9.48]	[9.45]	[9.57]	[5.53]	[3.40]
cma	-48.45	-29.66	-50.08	-44.56	-52.02	-59.90	-46.07
	[-7.10]	[-3.27]	[-5.58]	[-6.12]	[-5.02]	[-8.64]	[-4.60]
umd	1.01	1.97	1.48	-0.88	1.52	1.36	2.30
	[0.43]	[0.81]	[0.60]	[-0.35]	[0.61]	[0.57]	[1.00]
# months	684	684	684	680	684	684	680
$\bar{R}^2(\%)$	40	35	34	35	34	39	46

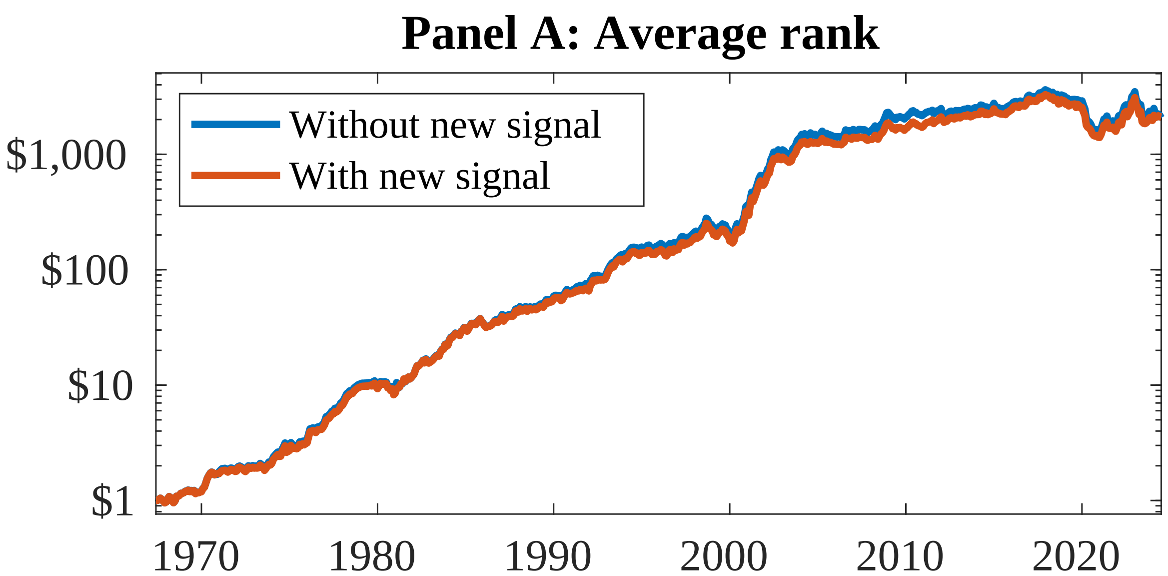


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as DTF. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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